Heritability Analysis of Complex Traits & Diseases:
Twin Study, Family Study, and GWAS

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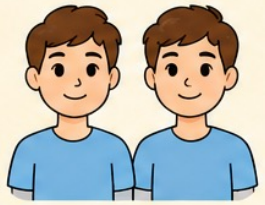
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ACE Model: Estimating Heritability of Height Using Twin Data

1 Twin study design

Compare height similarity between monozygotic (MZ) twins, who share 100% of their genes, and dizygotic (DZ) twins, who share on average 50% of their genes.

MZ twins
(100% shared genes)



More similar in height

DZ twins
(~50% shared genes)



Less similar in height

2 ACE model for height

An individual's height (H) is influenced by three sources:

$$H = A + C + E$$

A
(Additive genetic effects)



Effects of many genes that add up to influence height.

C
(Shared environment)



Environmental factors shared by twins (e.g., family background, nutrition).

E
(Unique environment)



Environmental factors unique to each individual (e.g., exercise, diet, random factors).

3 Expected twin correlations

Under the ACE model, the expected phenotypic correlations for height are:



MZ twins (100% shared genes)

$$r_{MZ} = A + C$$



DZ twins (~50% shared genes)

$$r_{DZ} = 0.5A + C$$

Note: The unique environmental component (E) does not contribute to the correlation between twins because it is unshared.

4 Estimating A and C from twin correlations

From the two equations,

$$r_{MZ} = A + C$$
$$r_{DZ} = 0.5A + C$$

we can solve for A and C:

$$A = 2(r_{MZ} - r_{DZ})$$
$$C = 2r_{DZ} - r_{MZ}$$

A: proportion of phenotypic variance due to additive genetic effects (heritability)

C: proportion due to shared environmental effects

5 Example: Height

Suppose a twin study finds the following height correlations:

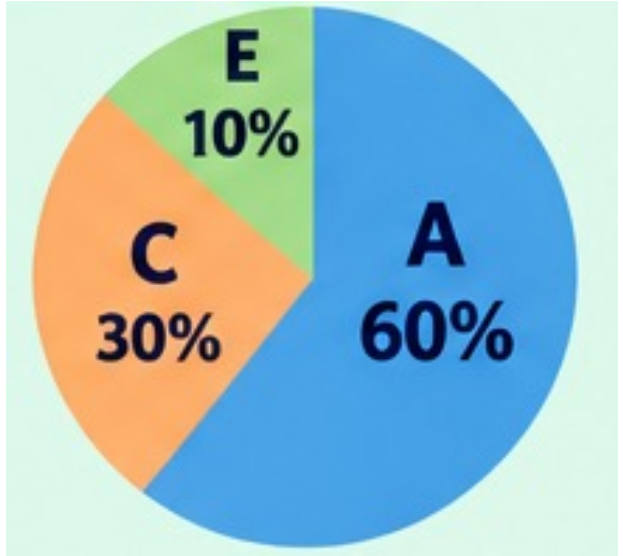
$$r_{MZ} = 0.90 \qquad r_{DZ} = 0.60$$

Then,

$$A = 2(0.90 - 0.60) = 0.60$$
$$C = 2(0.60) - 0.90 = 0.30$$

The remaining variance is due to unique environmental factors:

$$E = 1 - A - C = 1 - 0.60 - 0.30 = 0.10$$



General Phenotypic Variance Partitioning

1 Phenotypic model and heritability

$$H = A + D + C + E$$

An individual's height (H) is influenced by four components:

A — Additive genetic effects



Sum of effects of individual genetic variants (additive).

D — Dominance genetic effects



Interaction between gene pairs at the same locus (dominance).

C — Shared environment



Environmental factors that make family members similar (e.g., family background, nutrition).

E — Unique environment



Individual-specific environmental factors (e.g., exercise, diet, random factors).

Narrow-sense heritability (additive genetic variance)

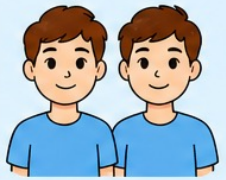
$$h^2 = \frac{A}{A+D+C+E}$$

Broad-sense heritability (total genetic variance)

$$H^2 = \frac{A+D}{A+D+C+E}$$

2 Expected twin correlations under the ADE/ACE model

Monozygotic (MZ) twins (100% shared genes)



Share 100% of both additive and dominance genetic effects, and shared environment.

Dizygotic (DZ) twins (~50% shared genes)



Share on average 50% of additive genetic effects, 25% of dominance effects, and shared environment.

Expected twin correlations

$$r_{MZ} = A + D + C$$

$$r_{DZ} = 0.5A + 0.25D + C$$

We have two equations but three unknown parameters (A, D, C), so they cannot be uniquely estimated without additional assumptions or data.

3 ACE model: assume no dominance ($D = 0$)

If dominance effects are negligible, we use the ACE model: $H = A + C + E$.

$$r_{MZ} = A + C$$
$$r_{DZ} = 0.5A + C$$

Solve for A and C

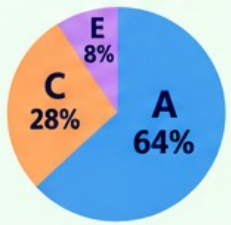
$$A = 2(r_{MZ} - r_{DZ})$$
$$C = 2r_{DZ} - r_{MZ}$$

Example: Height (ACE model)

Suppose $r_{MZ} = 0.92$ and $r_{DZ} = 0.60$. Then

$$A = 2(0.92 - 0.60) = 0.64$$
$$C = 2(0.60) - 0.92 = 0.28$$
$$E = 1 - A - C = 0.08$$

Variance components (ACE model)



4 ADE model: assume no shared environment ($C = 0$)

If shared environmental effects are negligible, we use the ADE model: $H = A + D + E$.

$$r_{MZ} = A + D$$
$$r_{DZ} = 0.5A + 0.25D$$

Solve for A and D

$$A = 4r_{DZ} - r_{MZ}$$
$$D = 2(r_{MZ} - 2r_{DZ})$$

Example: Height (ADE model)

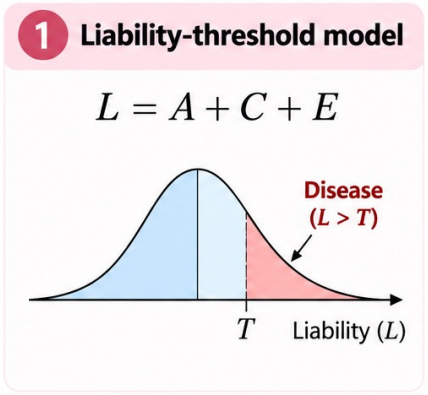
Suppose $r_{MZ} = 0.92$ and $r_{DZ} = 0.60$. Then

$$A = 4(0.60) - 0.92 = 1.48$$
$$D = 2(0.92 - 2 \times 0.60) = -0.56$$
$$E = 1 - A - D = 0.08$$

In this example, the estimate of D is negative, suggesting that the ADE model may not be appropriate for height (dominance effects are likely small).

Estimate Cancer Heritability in Nordic Twin Study: Liability Threshold mode + ACE

Diabetes: H1AC >6.5



Disease if $L > T$
Prevalence $K = P(L > T)$
Heritability assuming LT model

For MZ twins, we have likelihood

$$P(L_1 > T, L_2 > T|\rho_{MZ}) = P(L_2 > T|L_1 > T, \rho_{MZ})P(L_1 > T|\rho_{MZ})$$

Familial risk 0.196 **Prevalence 1.2%**

$$\rightarrow \rho_{MZ} = 0.60$$

$$\rightarrow \rho_{DZ} = 0.30$$

ACE: $h^2 = 2(\rho_{MZ} - \rho_{DZ}) = 2(0.6 - 0.3) = 0.6$

Population cumulative risk	1.2%
MZ twin familial risk (P(caner | co-twin has melanoma))	19.6%
DZ twin familial risk	6.1%

Data for melanoma

Mucci et al., JAMA, 2016

	Heritability	Shared Environment
Overall cancer	33 (30-37)	0
Head and neck	9 (0-60)	26 (0-65)
Stomach	22 (0-55)	6 (0-31)
Colon	15 (0-45)	16 (0-38)
Rectum and anus	14 (0-50)	10 (0-38)
Lung	18 (0-42)	24 (7-40)
Skin		
Melanoma	58 (43-73)	0
Nonmelanoma	43 (26-59)	0
Breast	31 (11-51)	16 (0-31)
Corpus uteri	27 (11-43)	0
Ovary	39 (23-55)	0
Prostate	57 (51-63)	0
Testis	37 (0-93)	24 (0-70)
Kidney	38 (21-55)	0
Bladder, other urinary organs	30 (0-67)	0

Mucci et al., JAMA, 2016

Variance Component Model for Heritability Estimate

Siblings: additive genetic effects only (no shared environment)

$$y_i = \mu + g_i + e_i$$

$$\mathrm{Var}(y_i) = \sigma_A^2 + \sigma_E^2$$

Siblings



Relatedness = 0.5

Expected covariance for siblings

$$\mathrm{Cov}(y_1, y_2) = 0.5\sigma_A^2$$

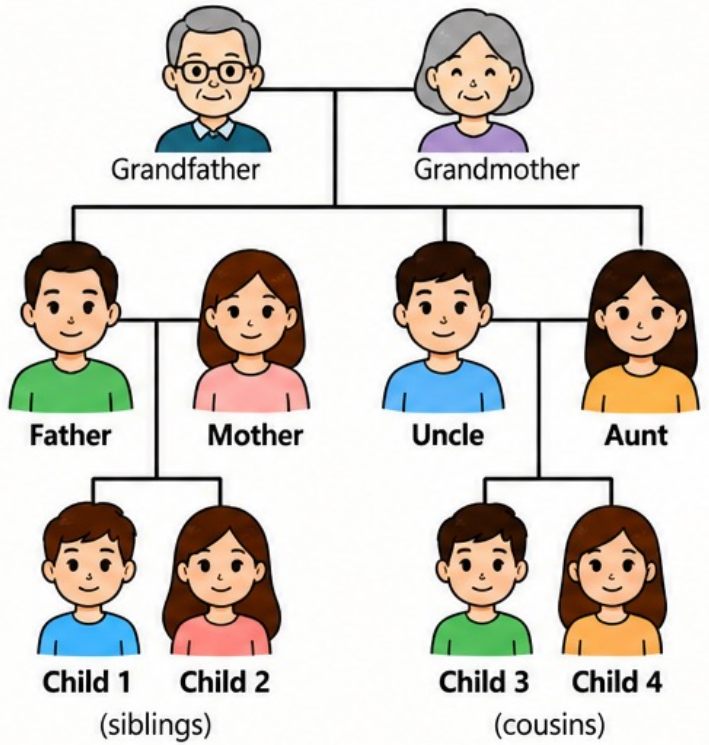
Extended families: variance component model using a genetic relatedness matrix

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{g} + \boldsymbol{e}$$

$$\boldsymbol{g} \sim \mathcal{N}(0, \sigma_A^2 \boldsymbol{K})$$

$$\boldsymbol{e} \sim \mathcal{N}(0, \sigma_E^2 \boldsymbol{I})$$

Extended family pedigree and genetic relatedness



Examples of expected genetic covariance ($\mathrm{Cov}(y_i, y_j) = 2\phi_{ij}\sigma_A^2$)

Relationship	Genetic relatedness ϕ_{ij}	$\mathrm{Cov}(y_i, y_j)$
Identical (same individual)	1	$2\sigma_A^2$
Parent – child	0.5	σ_A^2
Full siblings	0.5	σ_A^2
Grandparent – grandchild	0.25	$0.5\sigma_A^2$
Uncle/aunt – niece/nephew	0.25	$0.5\sigma_A^2$
First cousins	0.125	$0.25\sigma_A^2$

Heritability

$$h^2 = \frac{\sigma_A^2}{\sigma_A^2 + \sigma_E^2}$$

Missing Heritability Problem in GWAS

- High heritability from twins/families (80% height)
- Small phenotypic variance explained by identified SNPs (2% in 2007)
- Wrong estimate in twins/families or limited power of GWAS?
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Missing Heritability Problem in GWAS

- High heritability from twins/families (80% height)
- Small phenotypic variance explained by identified SNPs (<1% in 2007)
- Wrong estimate in twins/families or limited power of GWAS?
- An additive model in GWAS for T SNPs and N subjects

$$y_i = \sum_{t=1}^{T} \beta_t g_{it} + e_i$$

- When $N \to \infty$, we estimate β_t, and thus $h^2 = \frac{var(\sum_{t=1}^{T} \beta_t g_{it})}{var(y_i)}$.
- Can we estimate h^2 with limited sample sizes?
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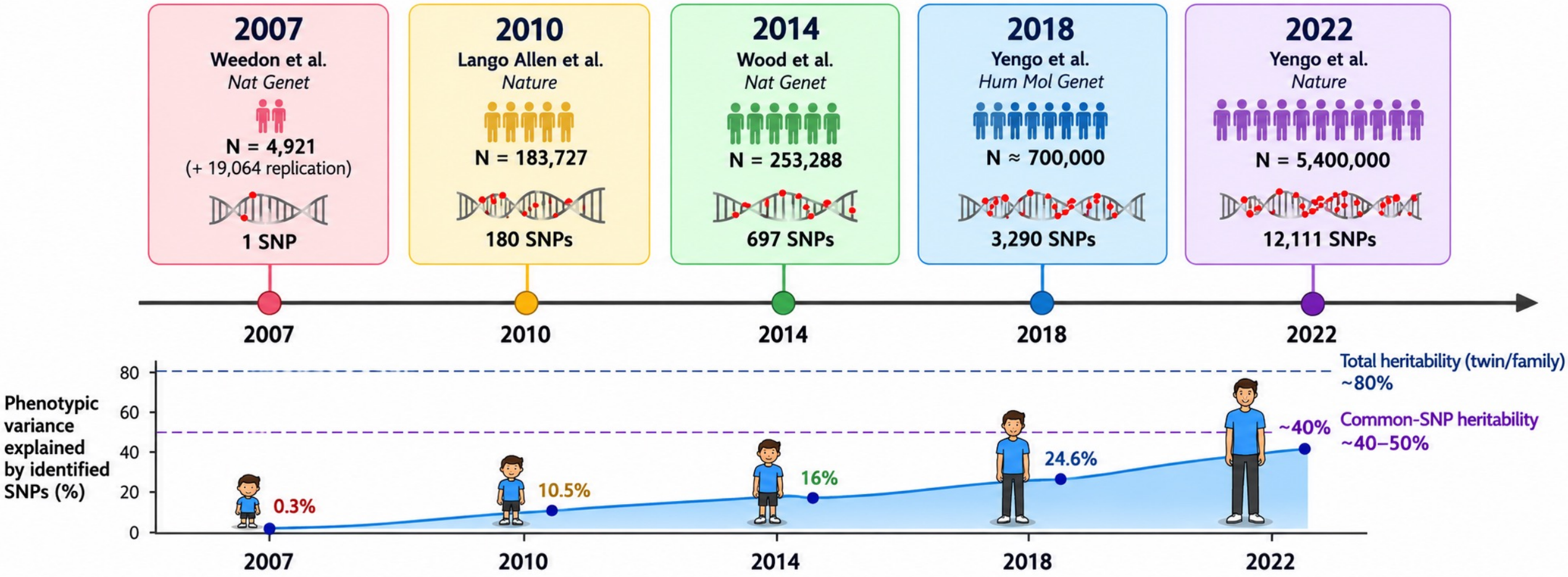
Estimate GWAS Chip Heritability

- GWAS Chip Heritability
  - Additive model $y_i = \sum_{t=1}^{T} \beta_t x_{it} + e_i$ for all SNPs in GWAS
  - x_{it} mean 0 and variance 1
  - Random effect model: $\beta_t \ i.i.d. \sim N(0, \sigma_g^2/T)$
  - Variance partitioning: $var(y_i) = \sigma_g^2 + \sigma_e^2$
  - Chip heritability $h^2 = \frac{\sigma_g^2}{\sigma_g^2+\sigma_e^2}$, phenotypic variance explained by all SNPs in GWAS
- Estimate h^2 using linear mixed model
  - Genetic similarity matrix G with $g_{ij} = \frac{1}{T}\sum_{t=1}^{T} x_{it}x_{jt}$.
  - Let $Y = (y_1, \cdots, y_N)'$, then $Y \sim MVN(0, \sigma_g^2 G + \sigma_e^2 I)$.
  - Average information REML can be used to estimate h^2.
- Height: 3925 unrelated subjects, 300,000 SNPs, $h^2 = 45\%$ $(s.e. = 7.5\%)$

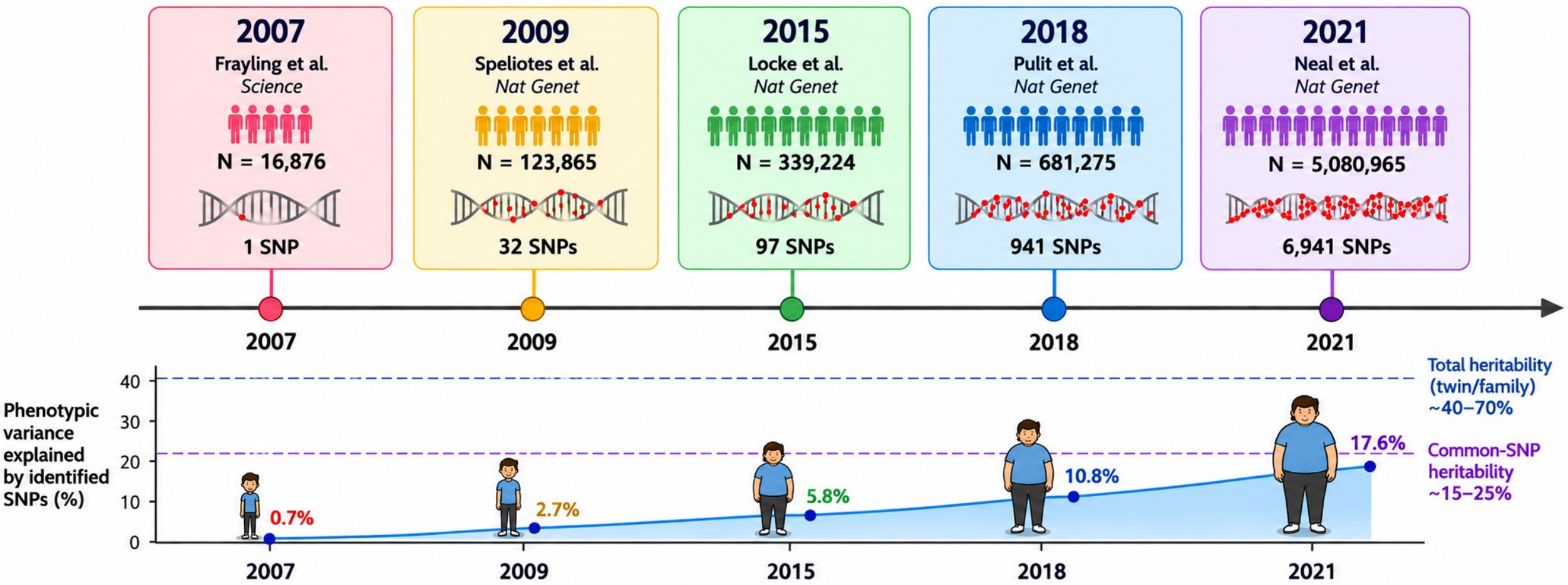
Yang et al., Nat. Genet. , 2010

Software: GCTA

GWAS of Human Height: Progression and Future



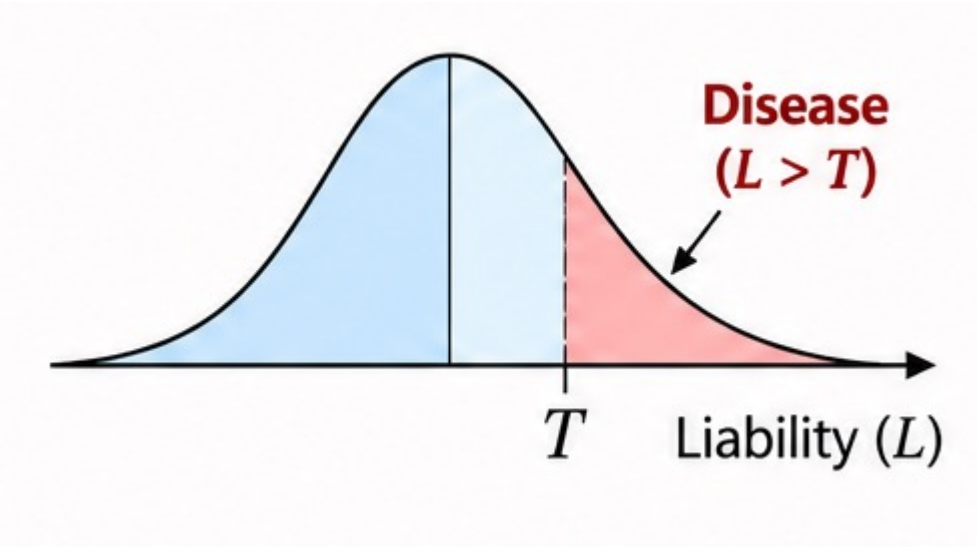
GWAS of Human BMI: Progression and Future



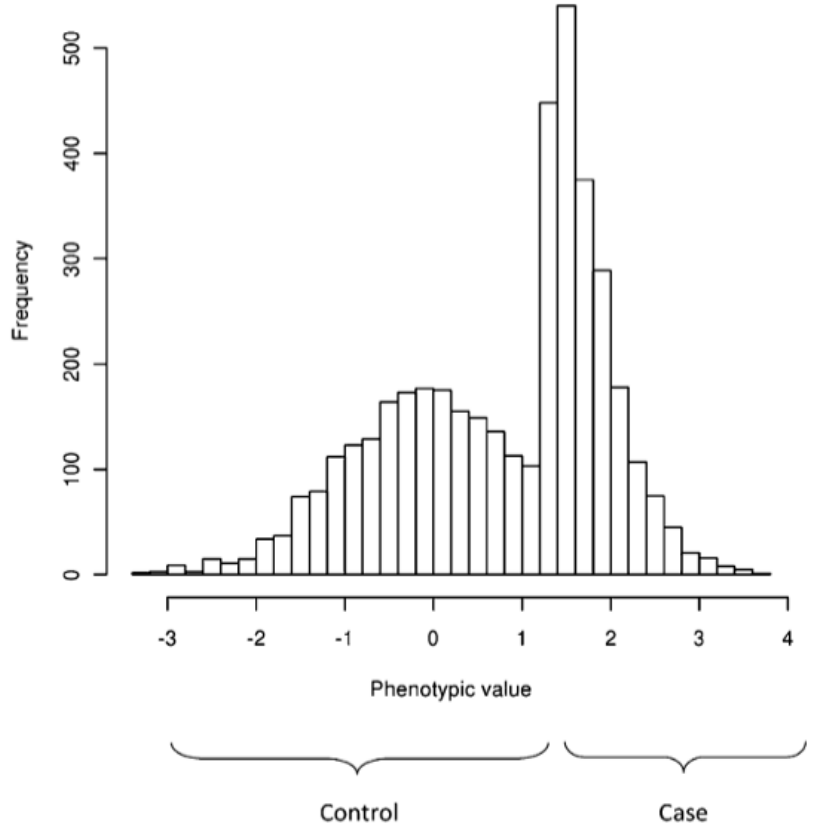
Additional Comments for Estimating Chip Heritability

- Require individual genotype data to calculate GRM
- At least 5000 subjects to have s.e. < 5%
- More SNPs included (e.g., imputed), higher chip heritability
- Refined models to allow effect size to depend on MAF
- When sample size is very large (e.g., UK BioBank), rare variants can be included into the linear mixed model.
  - N=350,000; MAF 0.01%-1%
  - Whole genome sequencing: common + rare variants ~ 88% family heritability
  - Wainschtein et al, Nature, 2026

Estimate GWAS Chip Heritability for Binary Diseases: Linear Mixed model + Liability Threshold model



$$P(y_i = 1) = P(L_i > T)$$
$$P(y_i = 0) = P(L_i < T)$$



Correction for oversampled cases

Software: GCTA

Yang et al., Nat. Genet. , 2010; Lett el at., AJHG, 2011, Yang et al., AJHG, 2011

Linear Mixed Model vs. LD Score Regression

- GCTA based on mixed model requires individual level data
- Modern GWAS are now typically based on meta-analysis, which collects summary statistics (beta/OR, s.e.) from studies.
- Can we estimate heritability using GWAS summary statistics and what's the price to pay?

Statistical Framework of LDSC

- Assuming $y = \sum_{k=1}^{M} \beta_k x_k + e$. Let $\hat{\beta}_k$ be marginal estimate, then

$$E\hat{\beta}_k = \sum_{i=1}^{M} r_{ki}\beta_i \quad \text{and} \quad Ez_k^2 = 1 + \sum_{i=1}^{M} r_{ki}^2 E\beta_i^2.$$

- Assuming $E\beta_i^2 = h_g^2/M$ be the heritability per SNP, then

$$Ez_k^2 = 1 + \frac{Nh_g^2}{M} l_k.$$

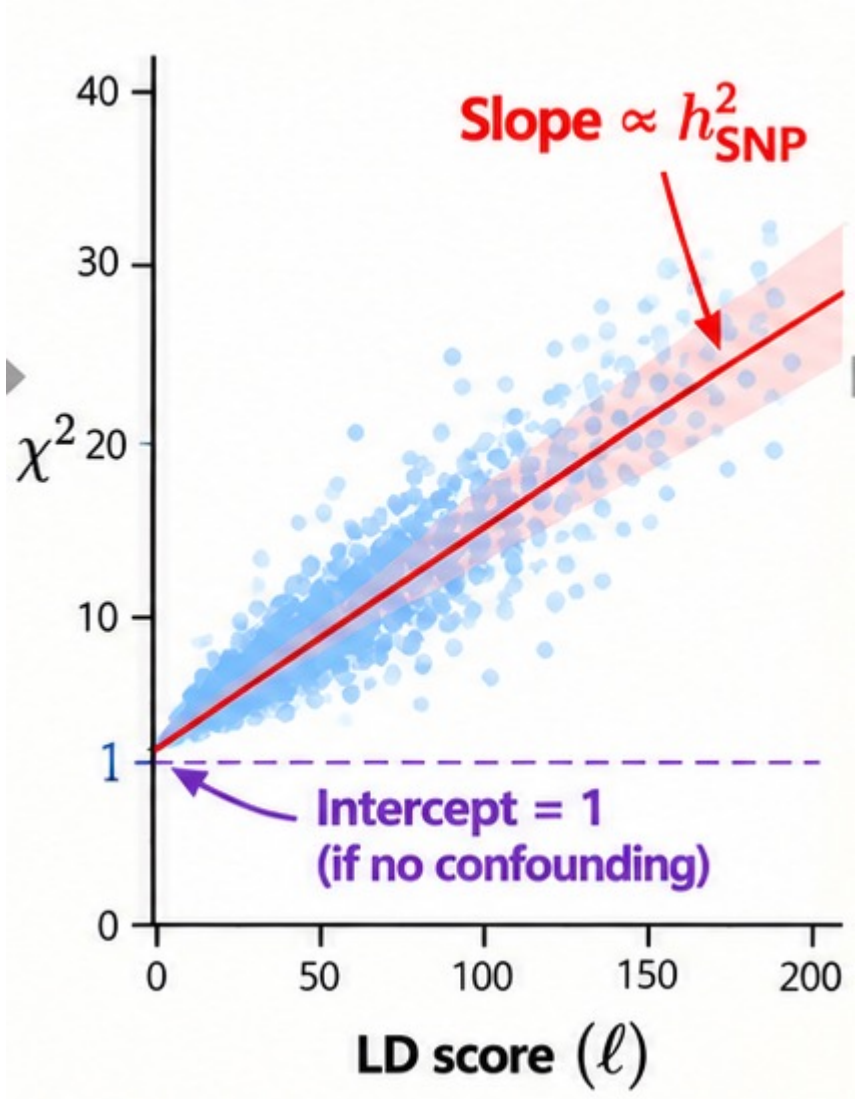
- $l_k = \sum_{i=1}^{M} r_{ki}^2$ is LD score of the variant and can be pre-calculated based on existing large scale refence data of the same ancestry.

Statistical Framework of LDSC

$$E(\chi_j^2) = 1 + \frac{N h_{SNP}^2}{M}\,\ell_j$$

χ_j^2	:	GWAS test statistic for SNP j
ℓ_j	:	LD score of SNP j
N	:	GWAS sample size
M	:	number of SNPs

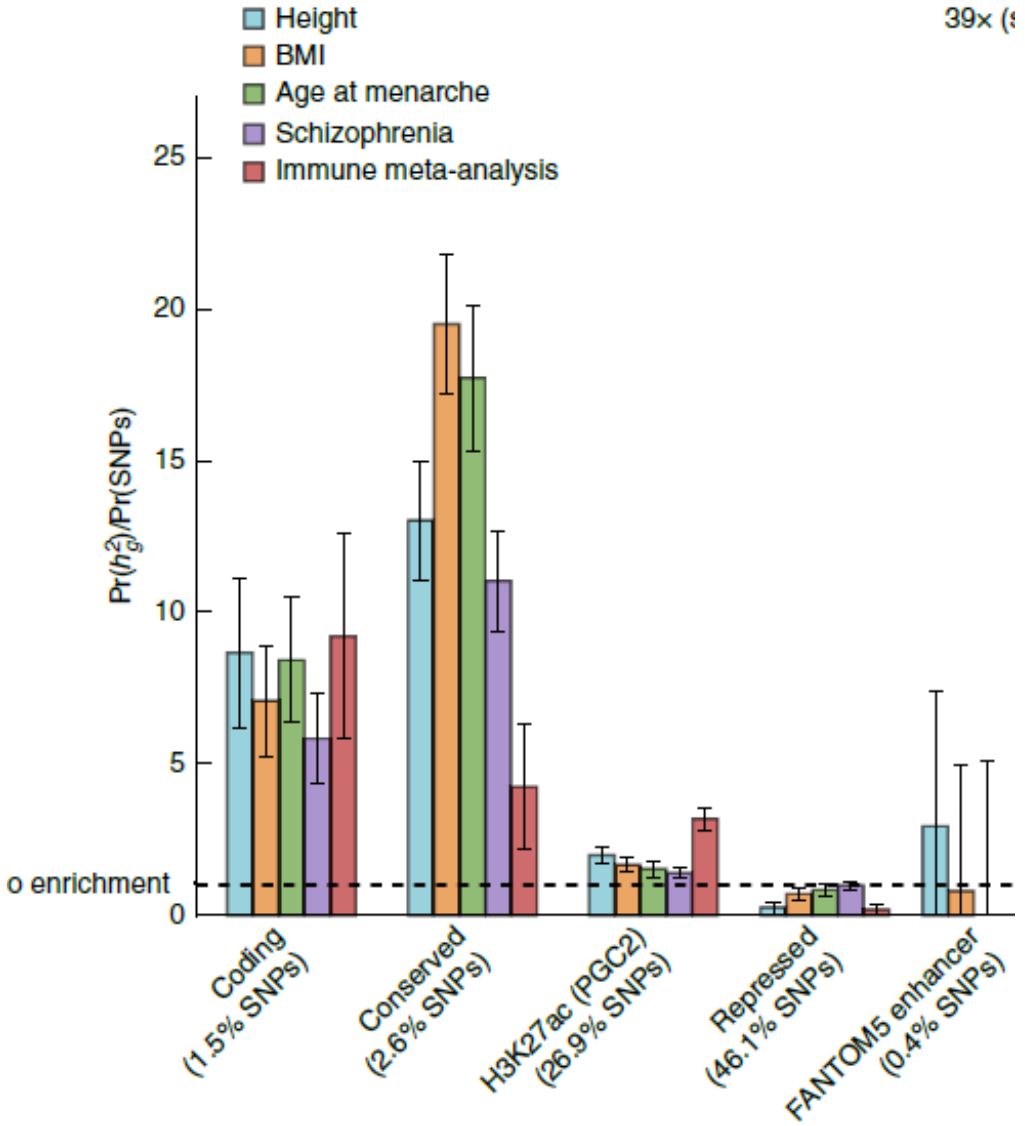
LD scores l_j can be pre-calculated for all variants



Bulik-Sullivan et al., NG, 2014; Finucane et al, NG, 2015

Heritability Enrichment Analysis Using LDSC

- Given a breast cancer GWAS (5M SNPs) and a DNA damage pathway (20K SNPs), is the pathway enriched for breast cancer risk?
- $\hat{h}^2_{overall}, \hat{h}^2_{DNA}. FC = \frac{\hat{h}^2_{DNA}}{\hat{h}^2_{overall}} \times \frac{5M}{20K}.$



Finucane et al, NG, 2015

Summary of Heritability Analysis

- Family-based heritability analysis includes additive, dominant effects, but may be inflated by shared environment effects. Small sample size is needed.
- GWAS chip heritability includes additive effects for common variants, may further include rare variants and dominant effects. Large sample size requires.
- LDSC uses summary statistics but loses about 70% statistical efficiency compared to GCTA.
- LDSC can infer population stratification!